

Track 2: Database Technologies

June 2026

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MR ROKESH TECHNOLOGY — Training & Internship Programs

Skill path: SQL/MySQL → PostgreSQL → MongoDB → Firebase → Design → CRUD → API Integration → Real-time DB

Related: [Master Syllabus](#) | [Duration Framework](#) | [Project Progression](#)

Track Overview

Who this is for: Aspiring DBAs, backend developers, data analysts, and full-stack developers needing database depth.

Prerequisites: Basic computer literacy. SQL familiarity helpful for 3-month+ tiers.

Tools & lab environment: MySQL Workbench, pgAdmin, MongoDB Compass, MongoDB Atlas (free tier), Firebase Console, DBeaver, Node.js/Python for API integration labs.

Certification targets (6–9 month): MySQL/PostgreSQL certification prep; MongoDB Associate Developer (optional)

Outcomes by Duration

Duration	Hours	Job Readiness	Key Deliverables
2 Weeks	40–50	SQL/NoSQL awareness	Library schema + connected app
1 Month	80–100	Junior DBA/analyst basics	Hybrid SQL + MongoDB project
3 Months	250–300	Internship-ready	Multi-tenant SaaS database design
6 Months	500–600	Production DBA-ready	Hospital/ERP hybrid schema capstone
9 Months	750–900	Industry-ready	Analytics platform + real-time backend

Weekly Rhythm (All Durations)

Mon–Thu labs | Fri project | Sat review | Sun GitHub README + schema documentation

2-Week Crash Course

Mini Project: Connect app to DB | **Level:** L1

Days	Topics	Lab	Deliverable	Self-Check
D1–2	RDBMS concepts, ER diagrams, normalization (1NF–3NF)	Design library schema	ER diagram + normalized tables	Explain 1NF, 2NF, 3NF with example
D3–4	MySQL: CREATE, SELECT, JOIN, GROUP BY, indexes	Student management DB	Working MySQL database + 10 queries	Write INNER vs LEFT JOIN
D5–6	PostgreSQL: advanced queries, window functions, CTEs	HR analytics queries	5 analytics queries with window functions	What is ROW_NUMBER()?
D7–8	MongoDB: collections, CRUD, aggregation pipeline	Blog/inventory NoSQL DB	MongoDB with 3 collections	When to choose MongoDB over SQL?
D9–10	API + DB integration (Node/Python), Firebase Realtime DB basics	Connect app to DB	Full-stack DB-connected app	Explain connection pooling basics

Submission checklist: - [] ER diagram submitted (draw.io or dbdiagram.io) - [] SQL scripts run without errors - [] MongoDB aggregation returns expected results - [] API endpoints read/write to database

1-Month Foundation

Capstone: Hybrid SQL orders + MongoDB catalog | **Level:** L2

Week	Focus	Project
W1	SQL fundamentals, MySQL admin, backup/restore basics	Library management system
W2	PostgreSQL, complex queries, indexing, EXPLAIN plans	E-commerce product DB
W3	MongoDB, Mongoose, schema design, validation	Blog platform database
W4	Firebase Realtime DB, security rules, hybrid architecture	Hybrid: SQL orders + MongoDB catalog

W4 acceptance criteria: - [] Orders in PostgreSQL/MySQL with referential integrity - [] Product catalog in MongoDB with flexible attributes - [] API layer queries both databases appropriately - [] Firebase demo for real-time inventory counter (optional stretch)

3-Month Job Ready

Capstone: Multi-tenant SaaS database design | **Level:** L3

Month	Topics	Projects
M1	SQL deep dive, ER modeling, normalization, MySQL + PostgreSQL admin	Library + Inventory DB
M2	Advanced SQL (CTEs, window functions), indexing, query optimization, EXPLAIN	Analytics dashboard queries
M3	MongoDB Atlas, aggregation pipelines, Redis caching, API integration, capstone	Multi-tenant SaaS database design

M3 Capstone acceptance criteria: - [] Tenant isolation strategy documented (schema-per-tenant or row-level) - [] Index strategy for top 5 queries documented - [] Migration scripts (up/down) included - [] Redis cache layer for read-heavy endpoints - [] Performance benchmark before/after indexing

6-Month Professional

Capstone: Hospital Management / ERP hybrid schema | **Level:** L4

Month	Topics	Projects
M1–M3	As 3-month track	Library, Inventory, Analytics projects
M4	Database administration: backups, replication, security, RBAC	DBA lab suite (5 labs)
M5	Real-time databases (Firestore), connectivity patterns, connection pooling	Real-time sync demo
M6	Performance tuning, sharding intro, capstone + cert prep	Hospital Management / ERP hybrid schema

M6 Capstone acceptance criteria: - ☐ Patient records, appointments, billing in SQL - ☐ Medical imaging metadata or logs in NoSQL - ☐ Backup/restore procedure documented and tested - ☐ Role-based DB access (doctor, admin, receptionist) - ☐ Query optimization report with EXPLAIN output

9-Month Advanced

Capstones: Analytics platform + Real-time chat/social app backend | **Level:** L5

Additional modules: Data warehousing basics, ETL pipelines (Python/Airflow intro), database migration strategies, multi-region replication.

Capstone 1 — Analytics Platform: - [] Star/snowflake schema design - [] ETL pipeline from operational DB to warehouse - [] 10+ analytical queries with documentation

Capstone 2 — Real-time Chat Backend: - [] Firebase or MongoDB Change Streams for real-time - [] Message indexing and pagination strategy - [] Load test with 100 concurrent connections

Assessment Rubric (Track-Specific)

Component	Weight	Criteria
Weekly Quiz	20%	SQL syntax, normalization, NoSQL concepts
Lab Completion	40%	Queries run; schemas valid; migrations applied
Project	30%	Schema design quality, optimization, documentation
Portfolio	10%	GitHub repos with ER diagrams, query files, README

Recommended Resources

- [MySQL Documentation](#)
- [PostgreSQL Tutorial](#)
- [MongoDB University \(Free\)](#)
- [Firebase Documentation](#)

Appendix: Mentor Notes

Common pitfalls: Skipping normalization; over-indexing; N+1 query patterns in API labs; storing JSON in SQL without justification.

Hardware: 8 GB RAM minimum; local MySQL + MongoDB or cloud free tiers.

Cross-track: [Track 1 — Full Stack](#) for API integration context; [Track 5 — AI/ML](#) for analytics overlap.